Dilip Sahu

B.Tech Computer Science & Engineering | NIT Mizoram | Backend & AI Engineering Internship
+91-8233958847 | bt24cs028@nitmz.ac.in | LinkedIn | GitHub | LeetCode | Codeforces | Portfolio | Kaggle

EDUCATION

National Institute of Technology Mizoram — Aizawl, Mizoram
B.Tech in Computer Science & Engineering / CGPA: ***7.98/10*** — *2024 – 2028*
Coursework: Distributed Systems, Cloud Computing, Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Artificial Intelligence, Machine Learning, OOP, Computer Organization & Architecture, Probability & Stochastics

Impulse Academy — Jaipur, Rajasthan
Class XII – RBSE / ***86%*** — *2023*

TECHNICAL SKILLS

Programming Languages: Python, **C/C++**, Java, JavaScript, **SQL**
Backend & Databases: FastAPI, **Node.js**, REST APIs, Server-Sent Events, Pydantic, **Redis**, **PostgreSQL**, MySQL, MongoDB, React, React Native
Distributed & Serverless: Serverless Deployment (Vercel), Distributed Systems, Cloud Computing, Message Queues (Redis FIFO), Fault Tolerance, High-Throughput Design
AI/ML: NLP, Deep Learning, Neural Networks, **Real-Time Inference**, **Multi-stage Transfer Learning**, Computer Vision (OpenCV, YOLO), **PyTorch**, **TensorFlow**
Algorithms: Data Structures & Algorithms, Graph Algorithms, Cryptography, Competitive Programming
Tools: Git, GitHub, Linux, VS Code | Quick learner of new technologies

EXPERIENCE

Zetheta Algorithms Pvt. Ltd. — Remote
Software Engineer Intern *(Project-Based)* — *Jul 2026 – Aug 2026*

- Built a **FastAPI reconciliation engine** for **100,000+ multi-bank transactions**; **RapidFuzz** entity matching
- Designed **automated data-ingestion pipelines** for **3+ banking formats** with **Pydantic** validation
- Implemented **tamper-evident PostgreSQL audit storage** (**SHA-256**); **100% transaction traceability**

PROJECTS

CyberGuard-AI – Real-Time Scam Detection GitHub — *Edge AI & NLP*

- Built an NLP scam-intent model with **multi-stage Transfer Learning**: frozen layers, then **discriminative fine-tuning** with layer-wise learning rates (avoids **catastrophic forgetting**)
- Delivered **real-time inference** using a **temporal sliding window** that keeps conversation context for live analysis
- **O(1) Bloom Filter** for instant known-caller and spam lookup

IoT-Based Bus-Tracking System Website | GitHub | Images — *Real-Time Data & Intelligent Systems*

- **Node.js** backend for **10,000+ req/sec** (5,000+ peak); **Redis** FIFO queue + **Server-Sent Events** for live updates
- **Haversine**-based ETA with **<50ms** map updates (Leaflet, PostgreSQL)

Coding & AI Club NIT Mizoram Website Website | GitHub — *Full-Stack & Serverless Deployment*

- Developed the full-stack site end-to-end and deployed it live on **Vercel (serverless platform)**
- Cut form-submission time from **2000ms to <100ms (95%)** using hover-intent prefetching via **DOM tree traversal**
- **Exponential backoff + jitter** for fault-tolerant, resilient API sync
- **Hash Set**-based duplicate check across multi-user team registrations in **O(N)** time

TourAI – Predictive AI Travel Planner GitHub — *Backend + Machine Learning*

- **FastAPI** backend serving model predictions, integrating ML output into a product feature
- **Prophet** demand forecasting (15-year Mizoram dataset); **Random Forest** for itinerary feasibility

ACHIEVEMENTS & LEADERSHIP

- **5G Innovation Hackathon 2026: Top 50 National Finalist**; CyberGuard-AI – **INR 1,00,000 seed funding**
- **Hackathons:** 1st Place – **3+ institute-level AI/innovation hackathons**; 1st Runner-Up – **Internal SIH**
- **Amazon ML Summer Challenge 2026: 95.29/100**; **Kaggle Global Rank 1,289** (Deep Past Language 2026)
- **Competitive Programming:** LeetCode + Codeforces + Striver's Sheet – **550+ problems**
- **Selective Programs:** Polaris Fellowship; HPCL-Mittal Energy Quest 3.0 (Pre-final stages)
- **Leadership:** Vice President – Coding & AI Club; Website Coordinator – T&P Office, NIT Mizoram
- **Certificates:** Cybersecurity (Zscaler); Drone Workshop (IIT Delhi & NIT Silchar); React Native (HackerRank)